Qian *et al. Cybersecurity* (2025) 8:21
https://doi.org/10.1186/s42400-024-00330-9

Cybersecurity

RESEARCH Open Access

Unveiling the veil: high-frequency components as the key to understanding medical DNNs' vulnerability to adversarial examples

Yaguan Qian[1], Renhui Tao[1], Huabin Du[1] and Bin Wang[2*]

Abstract

Deep Neural Networks (DNNs) have demonstrated outstanding performance in various medical image processing tasks. However, recent studies have revealed a heightened vulnerability of medical DNNs to adversarial attacks compared to their natural counterparts. In this work, we present a novel perspective by analyzing the disparities between medical datasets and natural datasets, specifically focusing on the dataset collection process. Our analysis uncovers unique differences in the data distribution across different image classes in medical datasets, a phenomenon absent in natural datasets. To gain deeper insights into medical datasets, we employ Fourier analysis tools to investigate medical DNNs. Intriguingly, we discover that high-frequency components in medical images exhibit stronger associations with corresponding labels compared to those in natural datasets. These high-frequency components distract the attention of medical DNNs, rendering them more susceptible to adversarial images. To mitigate this vulnerability, we propose a preprocessing technique called Removing High-frequency Components (RH) training. Our experimental results demonstrate that the application of RH training significantly enhances the robustness of medical DNNs against adversarial attacks. Notably, in certain scenarios, RH training even outperforms traditional adversarial training methods, particularly when subjected to black-box attacks.

Keywords Adversarial example, Fourier analysis, High-frequency component, Low-frequency component

Introduction

In recent years, Deep Neural Networks (DNNs) have gained significant popularity in the field of computer vision (CV) due to their exceptional performance. They have found extensive applications in various medical image processing tasks, including cancer diagnosis Bratchenko et al. (2022), diabetic retinopathy detection (Naz et al. 2022), pneumonia detection (Yu et al. 2021), organ localization (Gale et al. 2017; Navarro et al. 2020; Bejnordi et al. 2017; Xu et al. 2019; Roth et al. 2015). Notably, in 2018, the U.S. Food and Drug Administration (FDA) approved the first CV algorithm capable of medical diagnosis without human clinician input (Finlayson et al. 2018).

Despite the superior performance of DNNs and their potential to reduce medical costs, recent studies have highlighted their vulnerability to adversarial attacks (Ma et al. 2021; Koga and Takemoto 2022). These attacks can manipulate the final predictions by introducing imperceptible perturbations, thereby raising significant safety concerns regarding the deployment of DNNs in medical diagnosis systems (Finlayson et al. 2019).

*Correspondence:
Bin Wang
wangbin2@hikvision.com
[1] School of Science, Zhejiang University of Science and Technology, Hangzhou 3210023, Zhejiang, China
[2] Zhejiang Key Laboratory of Artificial Intelligence of Things (AIoT) Network and Data Security, Hangzhou 310051, Zhejiang, China

Springer Open

© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

Naturally, one may question why adversaries are interested in attacking DNNs in medical diagnosis systems. We identify two main reasons that could explain this phenomenon. First, the medical market is incredibly lucrative, making it an attractive target for attackers. By manipulating the results of medical diagnoses, adversaries can create false medical reimbursement claims, allowing them to illicitly profit from the system (Puttagunta et al. 2023). This poses a significant risk in terms of financial losses within the medical market. Second, adversarial attacks can be used to bias patient diagnoses, leading to false referrals or unnecessary prescriptions for medications (Bortsova et al. 2021). This not only has financial implications but also raises concerns about patient welfare and safety. Such situations can erode people's trust in Medical DNNs, ultimately limiting their widespread adoption in the medical field.

Medical robustness analysis

It is imperative to investigate the robustness of medical DNNs against adversarial attacks. Current studies have highlighted that medical DNNs (Abdou 2022) possess unique characteristics that distinguish them from natural DNNs, which primarily focus on classifying tasks involving natural images. Ma et al. (2021) have discovered that medical DNNs exhibit increased vulnerability compared to natural DNNs, as even slight perturbations can lead to severe medical malpractice. They attribute this fragility to the over-parameterization of DNNs. Both (Ma et al. 2021) and Yao et al. (2021) have observed that adversarial attacks tend to distort the deep representations of medical images to a greater extent. Additionally, Qi et al. (2021) have found that the variance of adversarial perturbations generated by medical DNNs is larger, rendering traditional attack methods more unstable. Furthermore, Bortsova et al. (2021) have determined that the robustness of medical DNNs is influenced by the initialization of model weights. However, these studies primarily concentrate on identifying vulnerabilities in medical DNNs at the model level.

Given the focus on the model level, several questions arise: (1) Do natural and medical datasets possess distinct properties at the data level? (2) Do medical datasets exhibit unique characteristics? Addressing these questions is of utmost importance to the research community as it can provide valuable insights into the fundamental differences between natural and medical datasets. Such understanding is crucial for advancing the field and developing effective strategies to enhance the robustness of medical DNNs against adversarial attacks.

Fourier domain analysis

This study utilizes the Fourier Transform as a valuable tool to investigate the underlying factors contributing to the heightened vulnerability of medical DNNs compared to their natural counterparts, focusing on the distinctive data distribution within medical image datasets. By decomposing the input signal into various frequencies, the Fourier Transform aids in discerning the disparities between medical and natural datasets. While Xu et al. (2019) suggest that DNNs exhibit a preference for fitting low-frequency information in training data, our research reveals that this observation does not entirely apply to medical datasets. Additionally, Wang et al. (2020) propose that low-frequency components are more closely associated with labels in natural datasets, but our findings challenge this notion. Furthermore, their assumption that incorporating high-frequency components enhances the generalization of convolutional neural networks is contradicted by our discovery that doing so can lead to counter-intuitive generalization behaviors, including the generation of adversarial images, ultimately compromising model robustness. Importantly, Maiya et al. (2021) conclude that the generation of adversarial images is not solely confined to high-frequency components but rather depends on the unique characteristics of the datasets themselves. By employing Fourier Transform to analyze the data distribution of medical images, our study aims to provide a comprehensive understanding of the heightened fragility of medical DNNs, contributing to the development of effective defense strategies and the enhancement of model robustness in medical diagnosis systems.

Our contributions

(1) We proposed that the composition of the data distribution of natural datasets and medical datasets are different, and the correlation between labels and images in medical datasets is not the "label"-"low-frequency components" like in natural datasets which is described in Wang et al. (2020).

(2) We analyzed the differences between different classes in medical datasets on data distribution. Contrary to natural datasets, we found these differences in high-frequency components are more important during data collection.

(3) We proposed removing high-frequency components (RH) training, a method of data preprocessing, to study the behavior of medical DNNs. We found that the high-frequency components, which are indispensable in data collection process, distract medical DNNs and impair their robustness. We also found that high-frequency components have less impact on clean accuracy

during training, which can be summarized as shown in Table 1.

(4) Our experiments also show that RH training can significantly improve the robustness of medical DNNs against both white and black-box attacks, which confirms that it is more effective in defending against adversarial attacks from the data perspective of medical datasets.

Fourier transform

For a matrix $m \in \mathbb{R}^{d_1 \times d_2}$, we denote its entries by $m[i, j]$, where $\forall i \in \{0, ..., d_1 - 1\}$, $\forall j \in \{0, ..., d_2 - 1\}$, the distance between its entries and its centroid is denoted by $d(z[i, j])$. We denote the 2D discrete Fourier transform (DFT) by $\mathcal{F} : \mathbb{R}^{d_1 \times d_2} \rightarrow \mathbb{C}^{d_1 \times d_2}$ and the inverse DFT by $\mathcal{F}^{-1}$. Given an image $x \in \mathbb{R}^{d_1 \times d_2}$, we denote its frequency component by $z = \mathcal{F}(x)$, correspondingly $x = \mathcal{F}^{-1}(z)$. We always shift the low-frequency components to the center of the spectrum z. If x has multiple channels, all procedure operates on each channel independently.

We use $z_l = \mathrm{L(z, r)}$ and $z_h = \mathrm{H(z, r)}$ to represent the low-frequency (LFC) and the high-frequency component (HFC) of z, where r is the radius of the filters. The low-frequency pass filter $\mathrm{L}(\cdot, \mathrm{r})$ and high-frequency pass filter $\mathrm{H}(\cdot, \mathrm{r})$ are formally defined as:

$$\begin{aligned} \mathrm{L(z, r)} &= \mathrm{z} * \underset{\mathrm{d}(\mathbb{1}[\mathrm{i,j}]) \leq \mathrm{r}}{\mathbb{1}} \\ \mathrm{H(z, r)} &= \mathrm{z} * \underset{\mathrm{d}(\mathbb{1}[\mathrm{i,j}]) > \mathrm{r}}{\mathbb{1}} \end{aligned} \tag{1}$$

where $\mathbb{1}$ and z have the same shape, the entries $\mathbb{1}[i, j]$ that satisfies the condition equals 1, otherwise, it equals 0. Thus we can decompose an image $x = \{x_l, x_h\}$, where $x_l = \mathcal{F}^{-1}(\mathrm{L(z, r)}) = \mathcal{F}^{-1}(\mathrm{z_l})$ and $x_h = \mathcal{F}^{-1}(\mathrm{H(z, r)}) = \mathcal{F}^{-1}(\mathrm{z_h})$, x_l and x_h are the low and high-frequency image of the original image x respectively. We decompose each image x_i in the C-class image dataset $\mathcal{D}$ into images formed by the low-frequency components and images formed by the high-frequency components, resulting in a low-frequency image dataset $\mathcal{D}_l$ and a high-frequency image dataset $\mathcal{D}_h$. Specifically, the low-frequency dataset $\mathcal{D}_l$ refers to an image dataset formed by retaining the low-frequency components and removing the high-frequency information from the images, while the high-frequency dataset $\mathcal{D}_h$ refers to an image dataset formed by retaining the high-frequency components and removing the low-frequency components from the images. $\mathcal{D} = \{\mathcal{D}_l, \mathcal{D}_h\} = \{(x_1, y_1), (x_2, y_2), \cdots\} = \{\{(x_{1l}, y_1), (x_{2l}, y_2), \cdots\}, \{(x_{1h}, y_1), (x_{2h}, y_2), \cdots\}\}$ where $\forall y_i \in \{1, \cdots, C\}$.

Generally speaking, the LFC in the image mainly includes the shape, outline and other information of the object, while the HFC mainly includes the detailed information such as texture and edge.

Table 1 The different importance of high-frequency components between medical and natural datasets

Datasets	Is high-frequency components More Important?	
	During data collection	During training
Medical	More	Less
Natural	Less	More

Assumptions

Inter-class classification tasks

In inter-class classification datasets, there are significant variations in the appearance or shape of images belonging to different classes, primarily due to the presence of low-frequency components. In the case of natural datasets like ILSVRC-2012 (Krizhevsky et al. 2012), the human eye heavily relies on these low-frequency components for accurate classification during the data collection process. As depicted in Fig. 1, even when the high-frequency components (HFC) of natural images (labeled as *TigerCat* and *Airliner*) are filtered out, human observers can still easily discern their respective classes. This observation suggests that the labels assigned to each class are primarily influenced by their low-frequency distributions. Furthermore, the impact of removing HFC during the data collection process is relatively minimal. Consequently, it can be inferred that the relationship between the data and labels can be understood as follows: $\mathcal{D} = \{\mathcal{D}_l, \mathcal{D}_h\} \approx \mathcal{D}_l = \{(x_{1l}, y_1), (x_{2l}, y_2), \cdots\}$, where $y_i \in \{1, \cdots, C\}$. Therefore, the data distribution mainly distinguished by DNNs is $\mathcal{L}^c$ where $c \in \{1, \cdots, C\}$.

Intra-class classification tasks

In the case of intra-class classification datasets, the visual appearance and shape of images belonging to different classes, such as *British Shorthair Cat* and *American Shorthair Cat*, exhibit striking similarities. Both classes are initially categorized as "Cat", and subsequently differentiated based on specific features unique to each breed, namely the high-frequency components. These distinct characteristics play a crucial role in discerning between *British* and *American Shorthair Cats*.

In the process of collecting data for intra-class classification datasets, the removal of HFC results in the recognition of both images as belonging to the general category of "cats". However, the identification of their specific classes necessitates the presence of additional HFC, highlighting the essential role of HFC in human visual perception for labeling images. In contrast to inter-class classification datasets, this data composition in intra-class classification datasets enhances the

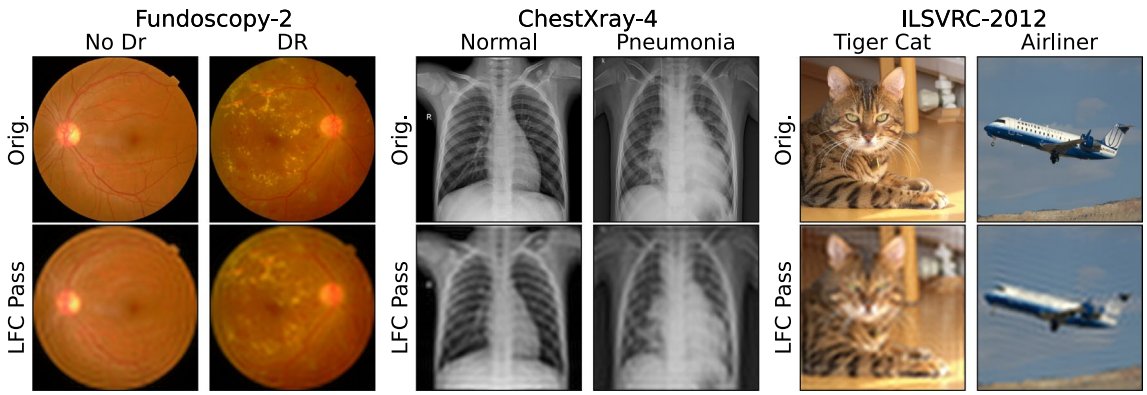

Fig. 1 Comparison of the original image and the low-frequency pass filtered image. All images are randomly selected from two classes of three datasets

correlation between HFC and the assigned labels. Consequently, the inclusion of HFC becomes crucial for accurately distinguishing between different classes within the same category.

In intra-class classification tasks, medical images and natural images share similar challenges and features, but they also exhibit differences. Medical images typically contain more complex biological textures, which are mainly reflected in the high-frequency components (HFC), making the role of HFC particularly critical in medical images. Similar to the subtle breed differences in natural images, such as *British Shorthair Cat* and *American Shorthair Cat*, intra-class classification in medical images also involves identifying small yet crucial details, such as variations in the textures of different types of lesions. In medical images, different categories (such as healthy versus diseased) may show high similarity in their low-frequency components (LFC), as the overall shape and structure of these images do not vary significantly. Therefore, HFC becomes the primary basis for distinguishing between these categories. While in natural images, high-frequency components are used to identify subtle differences between species or breeds, in medical images, they are key to distinguishing normal tissue from abnormal lesions. Details in these high-frequency components, such as edge sharpness and fine lesion textures, are essential for medical experts when annotating images and making diagnoses.

Moreover, due to the lack of relevant medical expertise, ordinary individuals can only rely on LFC for initial observation and struggle to capture the pathological features present in HFC. Only professionally trained doctors can effectively utilize the information in HFC for accurate diagnosis. This indicates that, although intra-class classification tasks are also present in natural images, the complexity and reliance on HFC in intra-class classification tasks in medical images are more pronounced. Furthermore, in the intra-class classification of medical images, HFC is not only crucial for improving the model's generalization ability but may also reduce the model's robustness due to excessive dependence on HFC. This is because detailed information in high-frequency components can be easily lost under noise interference or reduced imaging quality, leading to significant declines in model classification performance. Such phenomena are less common in natural images but more prevalent in medical images. Therefore, HFC is not only a key factor in distinguishing categories in intra-class classification but also, to some extent, a vulnerability of medical image classification models.

Consequently, we assume that for the image x^c in the intra-class classification dataset, the distribution $\mathcal{L}^c$ followed by its low-frequency image x^c_{il} across different classes is similar, while the primary difference lies in the distribution $\mathcal{H}^c$ followed by its high-frequency image x^c_{ih}. It should be noted that even though the $\mathcal{L}^c$ distributions of different classes are very similar, there still exist differences that can be identified by deep neural networks (DNNs). Therefore, the data distribution mainly distinguished by DNNs is $\mathcal{H}^c$, where $c \in \{1, \cdots, C\}$.

Nature of medical datasets

First, it is worth noting that a significant portion of medical datasets fall under the category of intra-class classification datasets. For instance, the Fundoscopy-2 dataset consists entirely of fundus images, while the Chest Xray-4 dataset comprises exclusively of chest X-rays. In these datasets, the primary variations between images of different classes lie in the types of lesions present. As illustrated in Fig. 1, it becomes evident that without HFC, the images from both medical datasets pose challenges for human observers in identifying the presence of lesions.

Second, it is important to note that individuals without relevant professional expertise can only perceive the LFC of medical images and are unable to capture the intricate details, namely the HFC. HFC, although not directly discernible to the human eye, can be perceived by relying on the information provided by LFC. Professionals, such as doctors, employ their expertise to identify specific texture details in the images and annotate them accordingly. This implies that even though natural datasets also encompass intra-class classification tasks, the nature of intra-class classification in medical datasets is more pronounced. Consequently, the relationship between HFC and labels in medical images exhibits a significantly higher correlation compared to that in natural images.

Furthermore, medical images exhibit a greater abundance of intricate biological textures, which are predominantly found in HFC. Building upon the characteristics of intra-class classification datasets and the findings from previous research (Wang et al. 2020), which suggest that HFC can enhance the generalization of DNNs but potentially compromise their robustness, we hypothesize that HFC serves as a crucial factor in rendering medical DNNs more susceptible to vulnerabilities.

Datasets and models

As shown in Table 2, we chose four medical datasets, including two binary medical datasets and two multi-class medical datasets, and two natural datasets: a subset of ILSVRC-2012 dataset (Krizhevsky et al. 2012) and CIFAR-10 dataset (Krizhevsky et al. 2009). Except for CIFAR-10, we splited the images randomly for training(80%) and testing(20%).

Medical datasets

Fundoscopy-2. From the Kaggle Fundoscopy Dataset (Kaggle 2015) we sampled a subset Fundoscopy-2, which includes 17,966 images, and the labels of each image are divided into five levels from "No DR" to "Mild / Moderate / Severe / Proliferative DR". We referred to Ma et al. (2021) and Gulshan et al. (2016) to divide this dataset into a binary dataset to distinguish non-referable (No DR and Mild) and referable (Moderate / Severe / Proliferative DR) diabetic retinopathy.

Chest Xray-2. We denoted the Kaggle Chest X-ray (Kaggle 2019) dataset as Chest Xray-2, which is a binary dataset for the pneumonia classification task and consists of 5,863 X-ray images labeled "pneumonia" or "normal".

Chest Xray-4. From the Kaggle Chest X-ray database (Wang et al. 2017) we selected the images labeled "mass" and "pneumothorax", and form a 4-classes dataset denoted by Chest Xray-4 with Chest Xray-2. Chest Xray-4 dataset includes 10,189 images.

ISIC-5. ISIC Challenge 2019 dataset (ISIC 2019) contains 8 classes of images, including 25,311 images. We removed the classes with few images and composed the remaining images into 5 classes (AK, BCC, BKL, MEL, NV) datasets, including 24,211 images.

Table 2 Number of the train/test images in each datasets

Datasets		Total	Train	Test
Medical datasets	Fundoscopy-2	17,966	14,372	3,594
	Chest Xray-2	5,840	4,672	1,168
	Chest Xray-4	10,189	8,152	2,037
	ISIC-5	25,331	20,265	5,066
Natural datasets	ImageNet-5	6,521	4,871	1,650
	CIFAR-10	60,000	50,000	10,000

Natural dataset

The natural datasets are generally huge dataset containing a large number of classes. For example, the ILSVRC-2012 (Krizhevsky et al. 2012) dataset contains 1,000 classes, the CIFAR-100(Krizhevsky et al. 2009) dataset contains 100 classes, but the number of classes in medical datasets is usually small. Therefore, in order to maintain the relative balance in the number of classes between the natural dataset and the medical dataset, we use CIFAR-10 and ImageNet-5 as the natural dataset for comparison.

ImageNet-5. We randomly selected five classes of images from ILSVRC-2012 (Krizhevsky et al. 2012) to form a new dataset. The number of classes and image size of this dataset are relatively consistent with the medical datasets we selected, which has good comparability.

CIFAR-10. The CIFAR-10 dataset (Canadian Institute For Advanced Research) Krizhevsky et al. (2009) is a collection of images that are commonly used to train machine learning and computer vision algorithms. The CIFAR-10 dataset contains 60,000 32×32 color images in ten different classes.

Model and weights

We opted to utilize the DenseNet121 model for our study, taking into account the findings of Paschali et al. (2018) that highlight the positive impact of dense blocks and skip connections on both the generalizability and robustness of medical DNNs. For the Fundoscopy-2 dataset, we performed a cropping operation to remove the black edges present in the original images. As for all four medical datasets and the ImageNet-5 dataset, we resized the images to dimensions of (256×256) and subsequently applied a center-crop operation to obtain images of size (224×224). However, for the CIFAR-10 dataset, we did not resize the images. During the training phase, we employed random rotation and random horizontal

flip as data augmentation techniques. Moreover, we utilized pre-trained weights obtained from the torchvision package for predictions on the ILSVRC-2012 (Krizhevsky et al. 2012) test dataset.

Removing HFC during training

To prove our assumption, we investigated the impact of high-frequency components in medical datasets on medical DNNs and propose the RH training method.

RH training setting

For all medical and natural datasets, we divided the train datasets into low-frequency and high-frequency components datasets $\{\mathcal{D}_l^{train}, \mathcal{D}_h^{train}\}$. We train the models on $\mathcal{D}_l^{train}$ with different radius r and test accuracies with the original test datasets, where $r \in \{5, 8, 10, 20, 30, 40, 60, 80, 112\}$, we denoted this training method as Removing HFC (RH) training. In the frequency domain, most of the energy is concentrated in LFC, which means that the amount of information contained in LFC is much larger than that in HFC. Therefore we chose a smaller interval in LFC and a larger interval in HFC. Since the image size of CIFAR-10 dataset is (32×32), we chose the radius $r \in \{2, 4, 6, 8, 10, 13, 16\}$. Note that while $r = 16$ for CIFAR-10 dataset and $r = 112$ for other datasets represent the clean training.

Generalization and HFC

Based on our initial assumption, we posited that the disparity between different classes in medical datasets primarily lies within the HFC. Consequently, we expected that DNNs would predominantly focus on learning the distribution $\mathcal{H}^c$. However, contrary to our expectations, the experimental results revealed an opposite trend. As depicted in Fig. 2, the test accuracies of medical DNNs exhibited a notable decline compared to natural DNNs as the radius r decreased. Specifically, the test accuracies of medical DNNs experienced a significant decrease only when r was less than 10. At this point, an excessive amount of information in the images was filtered out, resulting in limited availability of information. These findings indicate that the test accuracies of medical DNNs, trained without the presence of HFC, were less affected. Hence, it can be inferred that the generalization of medical DNNs is less reliant on the inclusion of HFC.

As depicted in Fig. 1, it becomes challenging for the human eye to discern the classes of two medical images once the HFC are removed. In contrast, DNNs exhibit remarkable discrimination capabilities based solely on the differences in the distribution $\mathcal{L}^c$ of each class. Presently, DNNs are primarily designed to classify large-scale natural datasets. Hence, we propose three potential reasons for their exceptional performance: (1) Most natural datasets are inter-class classification

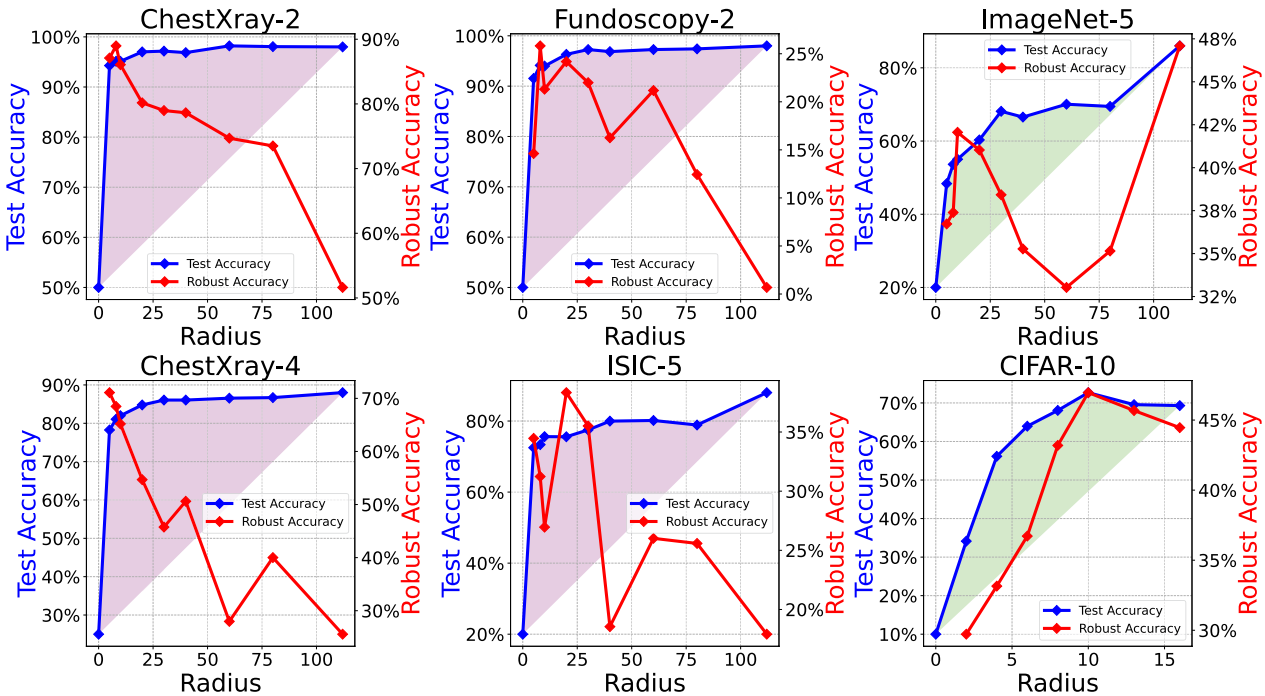

Fig. 2 Test accuracy and robust accuracy of RH trained models with different radius r. The blue line means the test accuracy, and the red line means the robust accuracy

datasets, as proposed by Wang et al. (2020) in their "natural selection" theory. This implies that the technologies developed within DNNs have been extensively selected by humans over time to better identify the LFC. As a result, DNNs naturally possess superior abilities in identifying LFC. (2) Medical datasets, in comparison to large-scale natural datasets, typically exhibit smaller data volumes and simpler data distributions. Consequently, DNNs find it easier to handle such datasets. (3) Ilyas et al. (2019) argue that the perceptual differences between DNNs and human vision may lead to unintuitive generalization behaviors of DNNs. DNNs have the ability to perceive data at a finer granularity than humans, capturing a wealth of content that the human eye cannot directly perceive. Therefore, even when a significant portion of HFC is removed during training, DNNs still retain high generalization capabilities.

Robustness and HFC

Furthermore, we employed the PGD method (Kurakin et al. 2018) with a perturbation budget of $\epsilon = 1/255$ to assess the robustness of each model. The results, depicted by the red line in Fig. 2, reveal interesting observations. Natural DNNs exhibit the highest level of robustness when trained on clean data, and removing HFC has a detrimental effect on their robustness. Conversely, as the radius r of robust HFC (RH) training decreases, the clean accuracies of medical DNNs experience only a slight decline, while their robust accuracies improve significantly. Notably, DNNs trained on clean data display the poorest level of robustness. This suggests that medical DNNs trained on datasets containing a greater amount of HFC exhibit lower levels of robustness.

This validates our hypothesis that the presence of HFC compromises the robustness of medical DNNs. DNNs demonstrate proficient classification performance on medical datasets solely utilizing the information contained in LFC, while HFC appears to be more akin to redundant information. When HFC is removed, DNNs are compelled to rely solely on LFC for association with labels. However, the labels of medical datasets exhibit a stronger correlation with HFC. Consequently, the inclusion of HFC may distract DNNs, potentially contributing to the heightened vulnerability of medical DNNs. Additionally, medical images often encompass intricate texture components, namely HFC, which are frequently inconsequential. Consequently, there may exist specific noise distributions within the $\mathcal{H}^c$ distribution, which can detrimentally impact the robustness of medical DNNs.

HFC distracting medical DNNs' attention

In this section, we delve into the reasons why HFC influences the robustness of medical DNNs when applied to medical images.

General noises in HFC

Medical images often contain various types of common and unavoidable noise. For instance, in ophthalmoscope or microscope images, the circular shape of the images results in large black borders surrounding the actual image region. The stark pixel value disparities between the real image area and the black border represent a form of HFC. Similarly, in Chest X-rays, noises such as watermarks labeled "L/R" are commonly employed to effectively distinguish between the left and right sides. These "L/R"-type noises are pervasive and can be observed in a substantial number of images, as illustrated in Fig. 4. Moreover, these noises can be easily eliminated using low pass filters, suggesting their association with HFC.

The specific data distribution of these HFC-related noises renders them easily detectable by DNNs. We regard these noises as resembling naturally embedded backdoor attacks, rather than artificial ones Li (2023), as they create a functional realm for potential adversarial attacks. During training, medical DNNs inadvertently learn to perceive these noises as significant features. Consequently, the attention of the DNNs is no longer "accurate," thereby rendering the medical DNNs more susceptible to vulnerabilities.

Attention of medical DNNs

To gain further insights into the behavior of medical DNNs, we employed Grad-CAM (Selvaraju et al. 2020) to identify the key regions in the input images that predominantly activate the network. The attention maps, as depicted in Fig. 3, illustrate the regions of interest for input images on RH-trained models with varying radius r. When trained on the natural dataset ImageNet-5, DNNs exhibit precise attention localization during clean training. However, as HFC is gradually removed, the attention areas of DNNs progressively deviate from the intended target. This observation explains the deteriorating robustness of natural DNNs.

Contrarily, in the Chest Xray-2 dataset, the primary attention area for clean-trained models is not the lesion area but rather the region containing the watermark in the upper left and lower right corners. This suggests that DNNs consider the watermark noise as a crucial feature of the image. Furthermore, RH-trained medical DNNs with larger r exhibit larger attention regions, wherein a minor perturbation at each pixel leads to a substantial output change. This observation further confirms that HFC in medical datasets distracts DNNs, rendering them

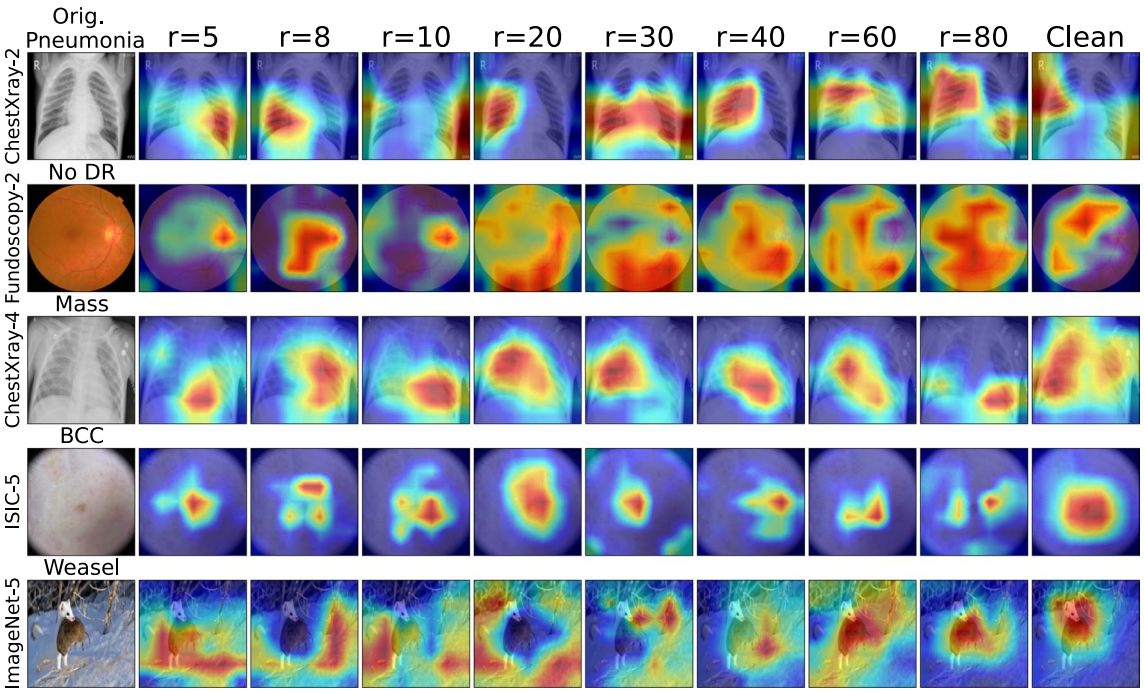

Fig. 3 Grad-CAM attention map of RH trained models with different radius *r*, where the redder regions indicate stronger attention

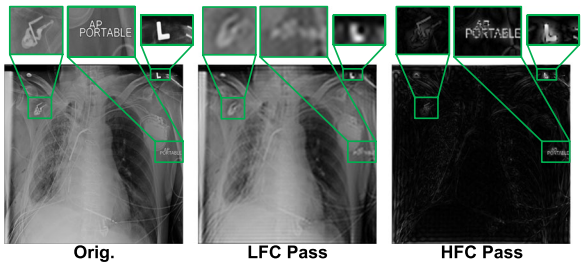

Fig. 4 Detail comparison of high-frequency and low-frequency pass filtered images

more vulnerable to adversarial attacks (Ma et al. 2021). Interestingly, we observed that RH-trained medical DNNs with smaller r display more accurate attention, focusing on the lesion area and exhibiting smaller attention regions. This finding also elucidates why removing HFC can significantly enhance robustness.

RH training against adversarial attacks

We compared the effects of RH training and adversarial training to further illustrate the impact of HFC.

Relative change in fourier transform

To observe the behaviors of medical DNNs against adversarial attacks in the frequency domain, we calculated the Relative Fourier Transform Change (RFC) of the clean input image x and its adversarial image $\tilde{x}$ as follows:

$$\mathrm{RFC} = \frac{1}{\mathrm{N}} \sum_{\mathrm{i}=1}^{\mathrm{N}} \left| \frac{\mathcal{F}(\tilde{\mathrm{x}}_\mathrm{i}) - \mathcal{F}(\mathrm{x}_\mathrm{i})}{\mathcal{F}(\mathrm{x}_\mathrm{i})} \right| \tag{2}$$

where $\mathcal{F}(\cdot)$ is the Fourier Transform, N is the number of images in the dataset, and $|\cdot|$ is the operation to take the absolute value. We calculated the average values of all images in the datasets as Eq. 2. RFC can measure which frequency component is more disturbed under adversarial attacks, i.e., the frequency component that is more vulnerable to DNNs.

RH training versus adversarial training

Performance of RH training. Adversarial training, recognized as the state-of-the-art defense method against adversarial attacks, leverages the weight information from the model's perspective. This approach can be viewed as an attempt to patch the vulnerabilities in DNNs. In contrast, RH training addresses the limitations inherent in the data itself. To investigate the efficacy of RH training, we conducted evaluations using PGD (Kurakin et al. 2018) and C&W (Carlini and Wagner 2017) attacks with a perturbation budget of $\epsilon = 8/255$. The obtained results, presented in Table 3, provide insights into the robustness of DNNs trained using different methods.

Table 3 Robustness comparison of clean, RH and adversarial trained models. We bolded the accuracies of RH training over clean training

Datasets	Training	Attack method		
		No attack	PGD	C&W
Fundoscopy-2	Clean	98.65	0.00	16.60
	RH.	96.31	24.18	30.12
	Adv.	96.31	91.80	93.31
Chest Xray-2	Clean	98.05	3.59	14.06
	RH.	95.13	88.97	75.04
	Adv.	97.01	92.73	91.63
Chest Xray-4	Clean	88.12	1.71	15.07
	RH.	78.25	71.08	68.92
	Adv.	82.72	74.37	75.26
ISIC-5	Clean	88.05	2.17	23.38
	RH.	75.57	38.33	26.11
	Adv.	76.45	62.43	62.86
ImageNet-5	Clean	86.00	20.24	36.60
	RH.	69.45	35.15	25.33
	Adv.	65.64	49.75	51.15
CIFAR-10	Clean	72.28	38.78	54.63
	RH.	63.96	34.47	32.04
	Adv.	68.67	56.42	56.21

The results demonstrate a notable enhancement in the robustness of all RH-trained medical DNNs compared to clean-trained DNNs when subjected to the strongest first-order attack PGD and optimization attack C&W. Meanwhile, the clean accuracies only exhibit a slight decrease. In both the Chest Xray-2 and Chest Xray-4 datasets, the performance of RH training is nearly on par with adversarial training, albeit with significantly lower time costs. However, the impact of RH training on natural datasets is less pronounced and can even undermine the robustness achieved through clean training.

This observation further reinforces the notion that HFC in medical datasets is a crucial factor that renders medical DNNs more susceptible to attacks. However, it is important to note that HFC is often indispensable during data collection as it enables human eyes to discern important features.

Similar behaviors between RH and adversarial training. Additionally, we observed that both RH training and adversarial training introduce similar modifications to medical DNNs. To quantify these changes, we computed the RFC (Relative Frequency Change) of clean, RH-trained, and adversarial-trained DNNs, as depicted in Fig. 6. Notably, for natural DNNs, RH training effectively reduces high-frequency perturbations. However, this reduction does not necessarily improve the robustness of natural DNNs. In contrast, adversarial training achieves robustness without heavily relying on the reduction of high-frequency perturbations. This discrepancy can be attributed to the generation of natural adversarial images, which may be less dependent on HFC. Consequently, the absence of this reliance on HFC may not be considered a flaw in natural datasets.

When medical DNNs are trained with clean examples, adversarial attacks tend to perturb the HFC of medical images to a greater extent. Both RH training and adversarial training effectively address this phenomenon by significantly reducing perturbations in the HFC. This indicates that the generation of medical adversarial images relies heavily on HFC. Consequently, the objective of adversarial training is to mitigate the damage caused by adversarial attacks on DNNs by suppressing high-frequency perturbations, akin to "Removing HFC". It is worth noting that RH training and adversarial training approach the suppression of high-frequency perturbations from different perspectives, yet both methods improve the robustness of DNNs, as evidenced in Table 3.

RH training against black-box attacks

We also conducted evaluations on the performance of RH-trained models against black-box attacks. Specifically, we employed PGD (Kurakin et al. 2018) with a perturbation budget of $\epsilon = 1/255$ for testing, using ResNet50 (He et al. 2016) and VGG16 (Simonyan and Zisserman 2015) as substitute models. The results, showcased in Fig. 5, demonstrate that under black-box attacks, the performance of RH training is comparable to that of adversarial training. In fact, in many instances, RH training achieves superior robustness compared to adversarial training. This finding suggests that medical datasets exhibit significant deficiencies in HFC, which affect DNNs trained on different architectures when learning from such data. Consequently, removing HFC can enhance the robustness against black-box attacks. This highlights the importance of comprehending medical datasets from a data perspective and devising appropriate methods to compensate for the data's shortcomings.

Discussion

This paper examines the medical dataset from a frequency domain perspective, but there are still several open problems that warrant discussion.

(1) As presented in Table 1, in contrast to natural datasets, the classification of medical images by human eyes relies more on HFC during the collection process, while DNNs rely less on HFC during training. This counter-intuitive behavior of DNNs contradicts the behavior of human eyes. Therefore, it is crucial to investigate the underlying reasons for this discrepancy.

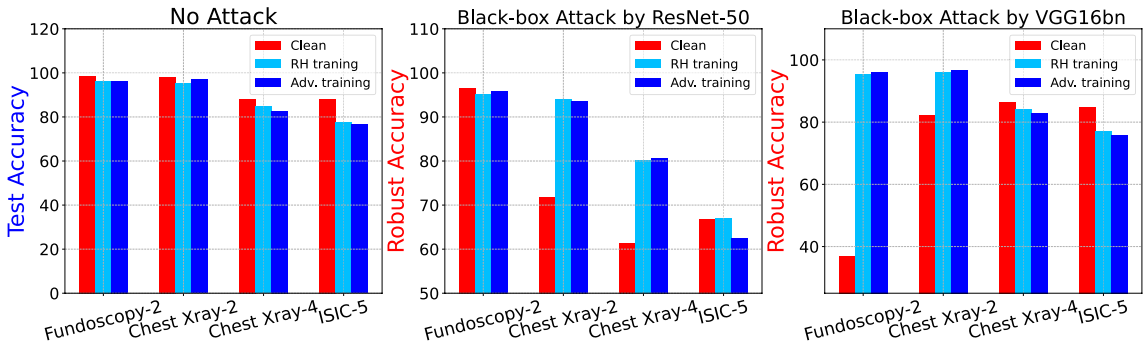

Fig. 5 Robustness comparison of clean, RH and adversarial trained DNNs against black-box attacks

(2) RH training significantly enhances the robustness of medical DNNs by focusing on the data perspective. However, it raises concerns about whether RH training, which exclusively utilizes LFC, may introduce new vulnerabilities to medical DNNs (Aguiar et al. 2023).

(3) Fig. 5 demonstrates that under black-box attacks, both adversarial and RH-trained DNNs are often less robust than standardly trained DNNs, which is a rarity in natural datasets. The "min-max" problem, typically addressed in adversarial training, is formulated as follows:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}}[\max_{\delta\in\mathcal{S}} L(\theta, x+\delta, y)] \tag{3}$$

where x denotes the original image, δ represents the perturbation, $\mathcal{S}$ denotes the set of possible perturbations, y is the correct label for the original image x, $\mathcal{D}$ represents the data distribution, and θ represents the model parameters. Does the fact that directly solving the "min-max" problem on the medical dataset leads to vulnerabilities imply a flaw in the approach?

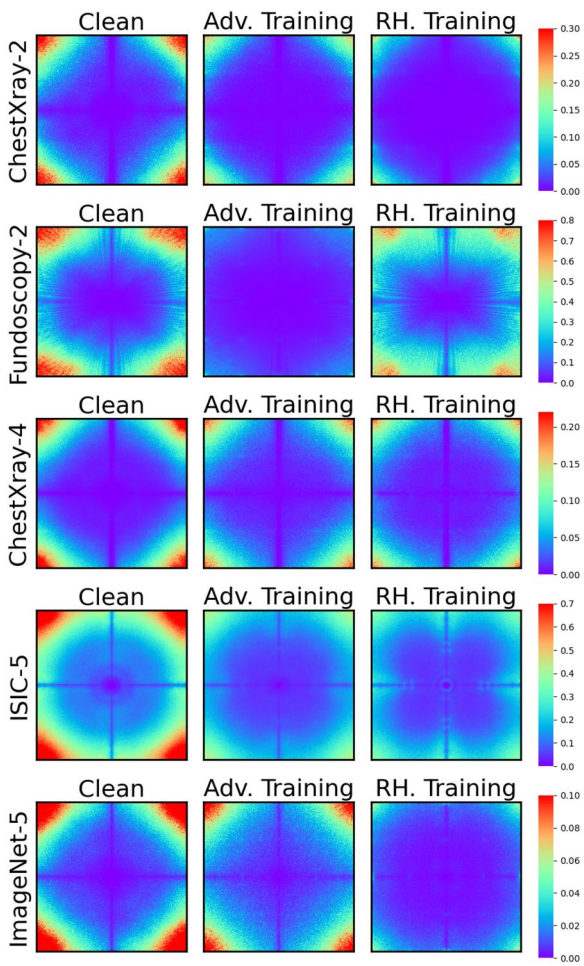

Fig. 6 RFC comparison of clean, RH and adversarial trained DNNs. The redder area indicates that the frequency domain is more disturbed, and we shifted the LFC to the center

Conclusions

This paper investigates the discrepancies in the frequency domain between medical datasets and natural datasets, with a particular focus on the distribution of HFC. Furthermore, we have observed that although HFC play a crucial role in human perception during data collection, they are relatively insignificant during the training of medical DNNs. Paradoxically, the presence of HFC can actually undermine the robustness of medical DNNs. Consequently, in the context of applying DNNs in the medical domain, addressing these "indispensable negative factors" becomes a pertinent question for discussion. Moreover, RH (Removing HFC) training techniques have been shown to significantly enhance the robustness of medical DNNs. Notably, RH training demonstrates superior effectiveness against black-box attacks, implying that data-level processing can bolster the overall robustness of medical DNNs. Moving forward, as mentioned in Sect. 8, we intend to further explore the robustness of medical DNNs from a data-oriented perspective.

Acknowledgements
The authors gratefully acknowledge the support from the National Natural Science Foundation of China under Grant 92167203 and Grant 62476250, the Zhejiang Provincial Natural Science Foundation of China under Grant LZ22F020007, and the Foundation of Zhejiang Key Laboratory of Multi-dimensional Perception Technology Application and Cybersecurity under Grant HIK2022008.

Author Contributions
Yaguan Qian: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Funding acquisition. Renhui Tao: Data curation, Investigation, Writing - Review & Editing. Huabin Du: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Software. Bin Wang: Writing - Review & Editing, Funding acquisition.

Funding
This work was supported by the National Natural Science Foundation of China under Grant 92167203 and Grant 62476250, Zhejiang Provincial Natural Science Foundation of China under Grant LZ22F020007, Foundation of Zhejiang Key Laboratory of Multi-dimensional Perception Technology Application and Cybersecurity under Grant HIK2022008.

Data Availability
The datasets used in this article are freely available on the internet. Detailed experimental data will be made available upon request.

Declarations

Ethics approval and consent to participate
All individual patient data appearing in this study are sourced from public datasets. All participants (or their legal guardians) have been informed about the purpose, methods, and risks of the study and have consented to participate.

Consent to publish
We confirmed that consent has been obtained from participants (or their legal guardians) for the individual patient data included in this study.

Competing interest
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Received: 10 May 2024 Accepted: 17 October 2024
Published online: 27 March 2025

References

Bratchenko IA, Bratchenko LA, Khristoforova YA, Moryatov AA, Kozlov SV, Zakharov VP (2022) Classification of skin cancer using convolutional neural networks analysis of raman spectra. Comput Methods Programs Biomed 219:106755
Naz H, Nijhawan R, Ahuja NJ (2022) An automated unsupervised deep learning-based approach for diabetic retinopathy detection. Medi Biol Eng Comput 60(12):3635–3654
Yu X, Wang S, Zhang Y (2021) Cgnet: A graph-knowledge embedded convolutional neural network for detection of pneumonia. In. Proc Manag 58(1):102411
Gale W, Oakden-Rayner L, Carneiro G, Bradley AP, Palmer LJ (2017) Detecting hip fractures with radiologist-level performance using deep neural networks. arXiv preprint arXiv:1711.06504
Navarro F, Sekuboyina A, Waldmannstetter D, Peeken JC, Combs SE, Menze BH (2020) Deep reinforcement learning for organ localization in CT. In: International Conference on Medical Imaging with Deep Learning, vol. 121, pp. 544–554
Bejnordi BE, Veta M, Van Diest PJ, Van Ginneken B, Karssemeijer N, Litjens G, Van Der Laak JA, Hermsen M, Manson QF, Balkenhol M et al (2017) Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer. Jama 318(22):2199–2210
Xu X, Zhou F, Liu B, Bai X (2019) Multiple organ localization in CT image using triple-branch fully convolutional networks. IEEE Access 7:98083–98093
Roth HR, Lu L, Farag A, Shin H-C, Liu J, Turkbey EB, Summers RM (2015) Deeporgan: Multi-level deep convolutional networks for automated pancreas segmentation. In: International Conference on Medical Image Computing and Computer-assisted Intervention, pp. 556–564
Finlayson SG, Chung HW, Kohane IS, Beam AL (2018) Adversarial attacks against medical deep learning systems. arXiv preprint arXiv:1804.05296
Ma X, Niu Y, Gu L, Wang Y, Zhao Y, Bailey J, Lu F (2021) Understanding adversarial attacks on deep learning based medical image analysis systems. Pattern Recognit. 110:107332
Koga K, Takemoto K (2022) Simple black-box universal adversarial attacks on deep neural networks for medical image classification. Algorithms, 144
Finlayson SG, Bowers JD, Ito J, Zittrain JL, Beam AL, Kohane IS (2019) Adversarial attacks on medical machine learning. Science 363(6433):1287–1289
Puttagunta M, Subban R, Babu CNK (2023) Adversarial examples: attacks and defences on medical deep learning systems. Multim Tools Appl 82(22):33773–33809
Bortsova G, González-Gonzalo C, Wetstein SC, Dubost F, Katramados I, Hogeweg L, Liefers B, Ginneken B, Pluim JP, Veta M et al (2021) Adversarial attack vulnerability of medical image analysis systems: Unexplored factors. Med Image Anal 73:102141
Abdou MA (2022) Literature review: efficient deep neural networks techniques for medical image analysis. Neural Comput. Appl., 5791–5812
Ma X, Niu Y, Gu L, Wang Y, Zhao Y, Bailey J, Lu F (2021) Understanding adversarial attacks on deep learning based medical image analysis systems. Pattern Recognit 110:107332
Yao Q, He Z, Lin Y, Ma K, Zheng Y, Zhou SK (2021) A hierarchical feature constraint to camouflage medical adversarial attacks. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 36–47
Qi G, Gong L, Song Y, Ma K, Zheng Y (2021) Stabilized medical image attacks. In: 9th International Conference on Learning Representations
Xu Z-QJ, Zhang Y, Luo T, Xiao Y, Ma Z (2019) Frequency principle: Fourier analysis sheds light on deep neural networks. arXiv preprint arXiv:1901.06523
Wang H, Wu X, Huang Z, Xing EP (2020) High-frequency component helps explain the generalization of convolutional neural networks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 8684–8694
Maiya SR, Ehrlich M, Agarwal V, Lim S-N, Goldstein T, Shrivastava A (2021) A frequency perspective of adversarial robustness. arXiv preprint arXiv:2111.00861
Krizhevsky A, Sutskever I, Hinton GE (2012) Imagenet classification with deep convolutional neural networks. Advances in neural information processing systems **25**
Krizhevsky A, Hinton G, et al (2009) Learning multiple layers of features from tiny images. Handbook of Systemic Autoimmune Diseases **1**(4)
Kaggle (2015) Kaggle diabetic retinopathy detection challenge. https://www.kaggle.com/c/diabetic-retinopathy-detection
Gulshan V, Peng L, Coram M, Stumpe MC, Wu D, Narayanaswamy A, Venugopalan S, Widner K, Madams T, Cuadros J, et al (2016) Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. Jama, 2402–2410
Kaggle (2019) Chest X-Ray Images (Pneumonia), 2019. https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia
Wang X, Peng Y, Lu L, Lu Z, Bagheri M, Summers RM (2017) Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2097–2106
ISIC (2019) The international skin imaging collaboration. https://www.isic-archive.com
Paschali M, Conjeti S, Navarro F, Navab N (2018) Generalizability vs. robustness: adversarial examples for medical imaging. arXiv preprint arXiv:1804.00504
Ilyas A, Santurkar S, Tsipras D, Engstrom L, Tran B, Madry A (2019) Adversarial examples are not bugs, they are features. Advances in neural information processing systems **32**

Kurakin A, Goodfellow IJ, Bengio S (2018) Adversarial examples in the physical world. In: Artificial Intelligence Safety and Security, pp. 99–112
Li Y (2023) Poisoning-based backdoor attacks in computer vision. In: Thirty-Seventh AAAI Conference on Artificial Intelligence,, pp. 16121–16122
Selvaraju RR, Cogswell M, Das A, Vedantam R, Parikh D, Batra D (2020) Grad-cam: Visual explanations from deep networks via gradient-based localization. Int J Comput Vis 128:336–359
Carlini N, Wagner D (2017) Towards evaluating the robustness of neural networks. In: 2017 Ieee Symposium on Security and Privacy (sp), pp. 39–57
He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 770–778
Simonyan K, Zisserman A (2015) Very deep convolutional networks for large-scale image recognition. In: 3rd International Conference on Learning Representations, ICLR
Aguiar EJD, Costa MVL, Traina C, Traina AJM (2023) Assessing vulnerabilities of deep learning explainability in medical image analysis under adversarial settings. In: International Symposium on Computer-Based Medical Systems, pp. 13–16

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.